

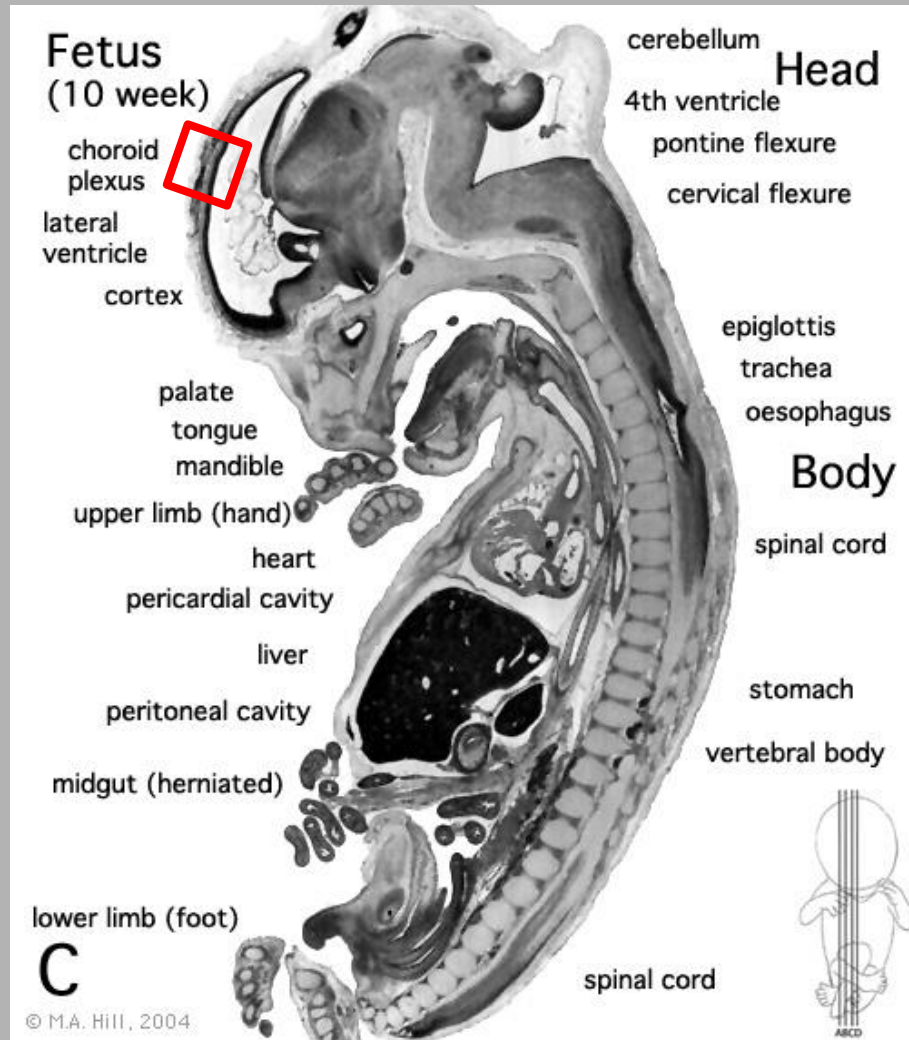
# Session Objectives

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- Students will be able to describe the basic mechanisms of neuronal proliferation in the fetal ventricular zones and how this relates to neocortical size and disorders such as microcephaly.
- Students will be able to describe the basic principles of neuronal migration in the fetal ventricular zone, subventricular zone, and cortical plate and how this relates to the establishment of the neocortical lamina.
- Students will be able to describe the mechanisms underlying disorders of neuronal migration such as subcortical band heterotopia, lissencephaly, etc.
- Students will be able to describe the basic organization of the formation of sulci and gyri in the developing neocortex.

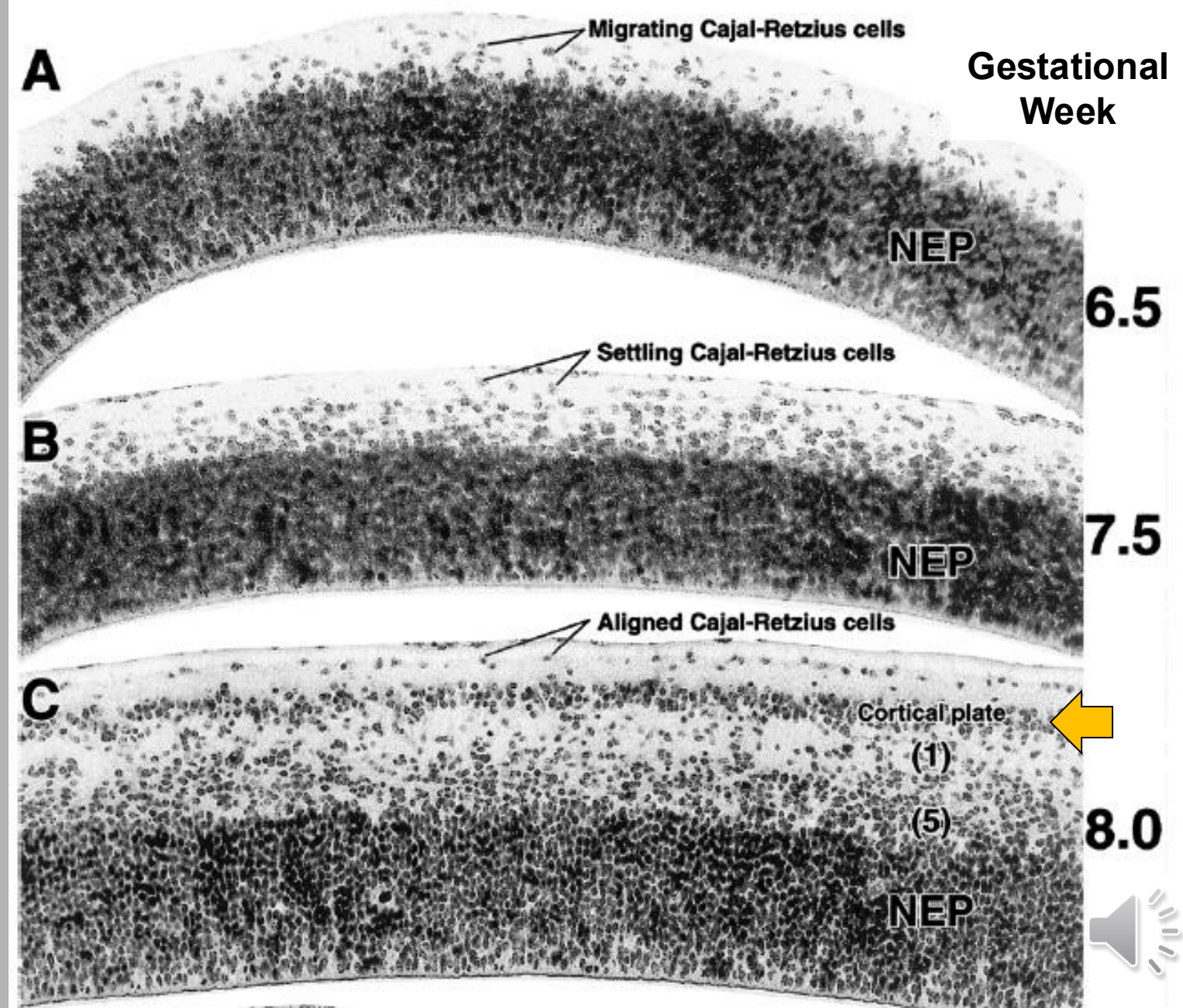


# Neuronal Production - making and keeping the right #'s of neurons



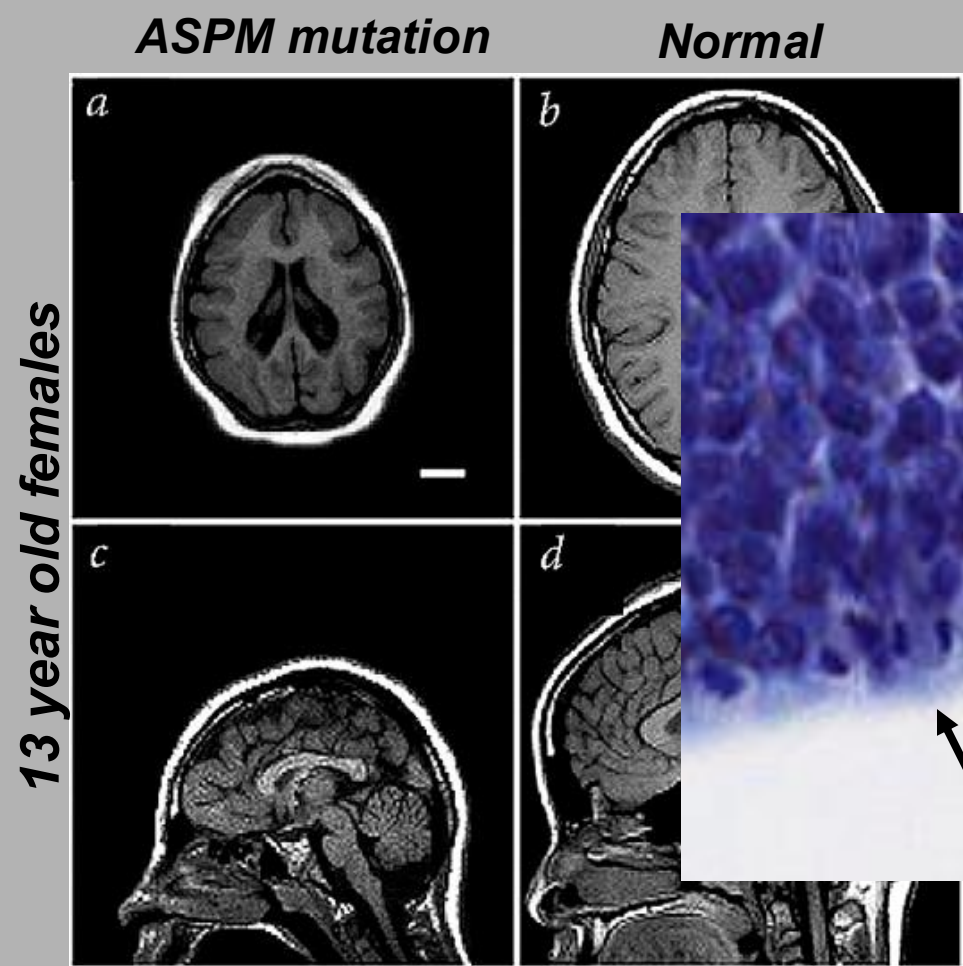
## Neocortical development

- nearly all cortical neurons are made prenatally
- all excitatory neurons are made at the VZ

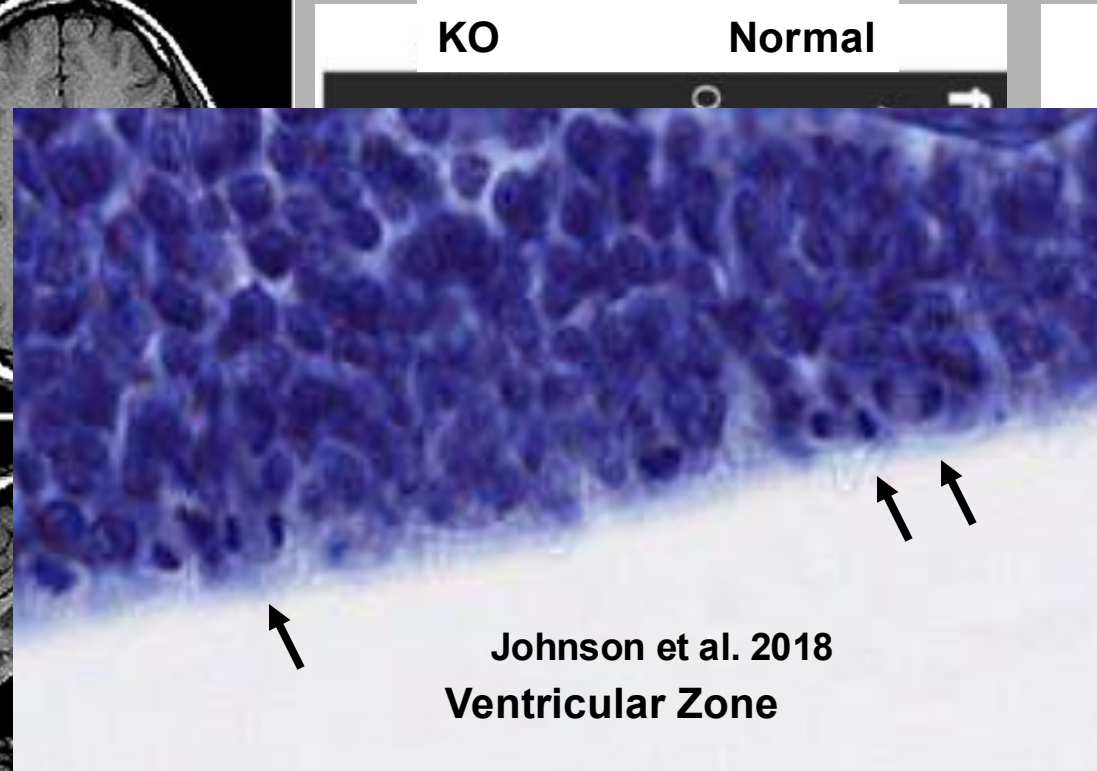


# Why is it important to understand neuronal proliferation in the brain?

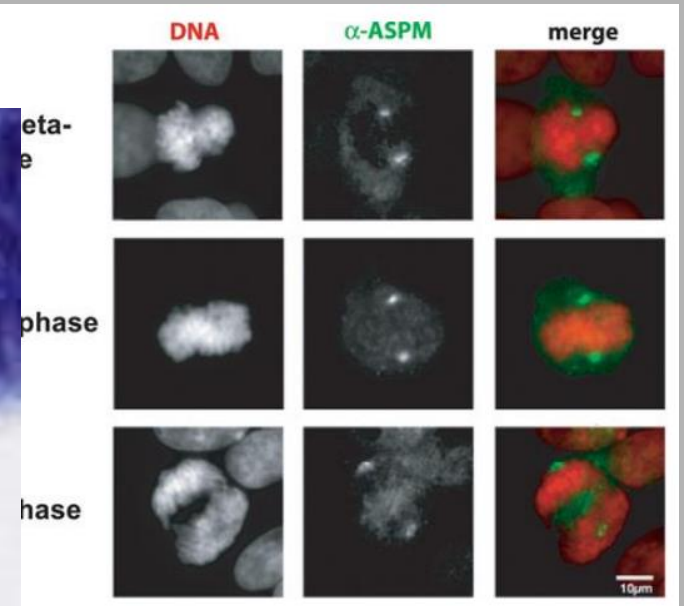
## Neuronal production disorders: *primary microcephaly*



Bond et al. 2002



\* ASPM important for mitotic spindle formation



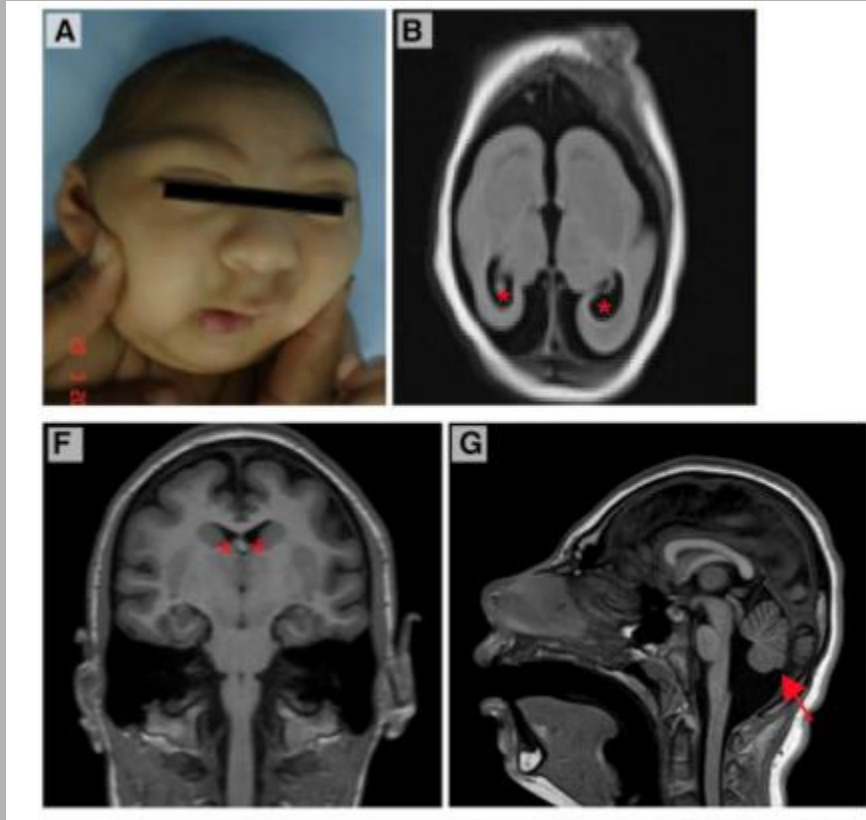
Kouprina et al. 2005



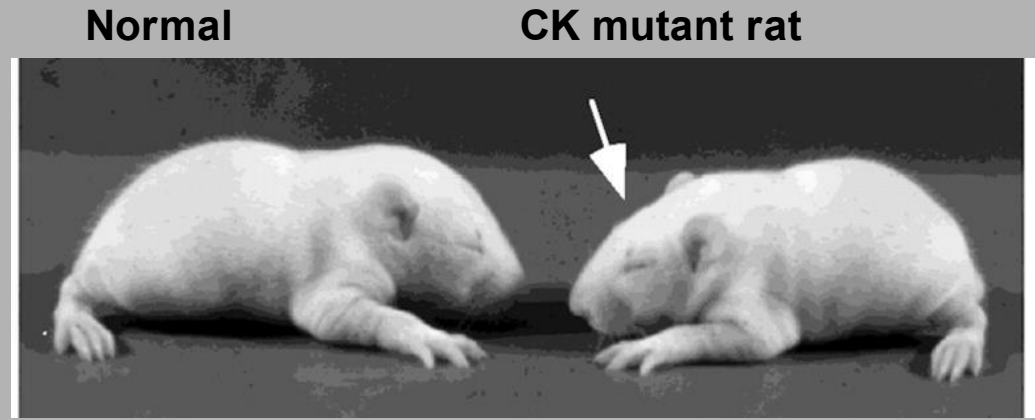
# Why is it important to understand neuronal proliferation in the brain?

## Neuronal production disorders: *primary microcephaly*

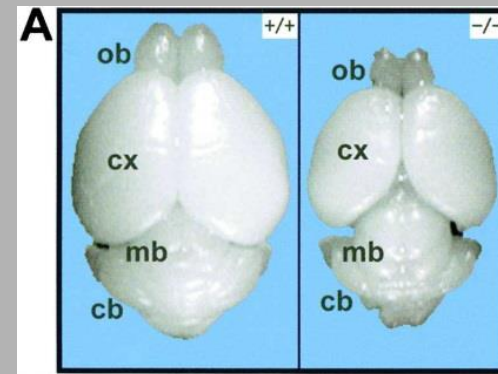
- \* Citron kinase (CK) mutations cause microcephaly in humans, rats, and mice



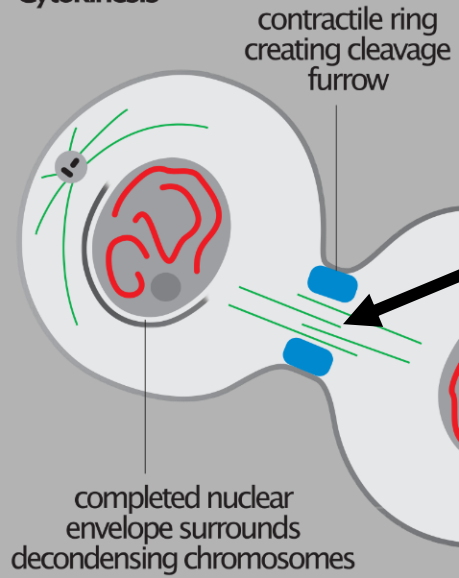
Harding BN, et al. Am J Hum Genet. 2016 Aug 4;99(2):511-20.  
Roberts et al. 2000; J Neuroscience  
DiCunzio et al. 2000; Neuron



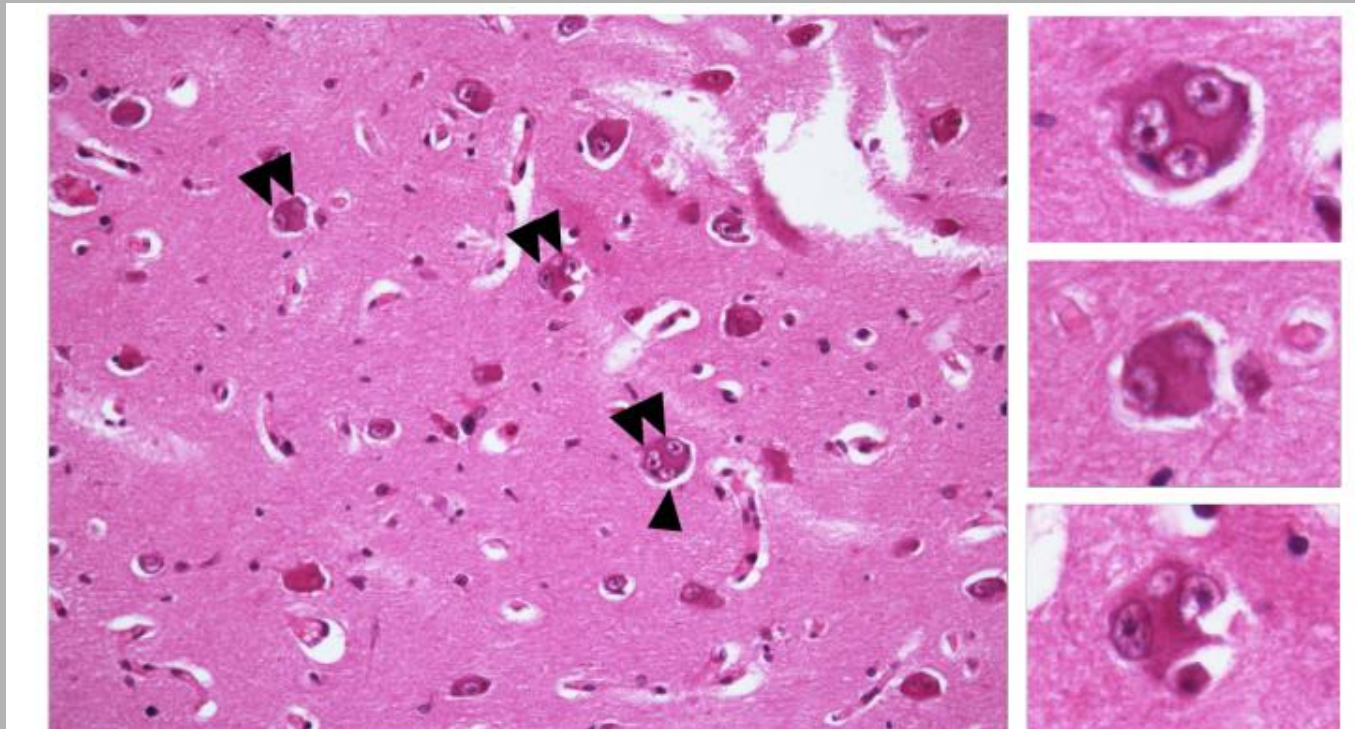
Normal CK KO mouse



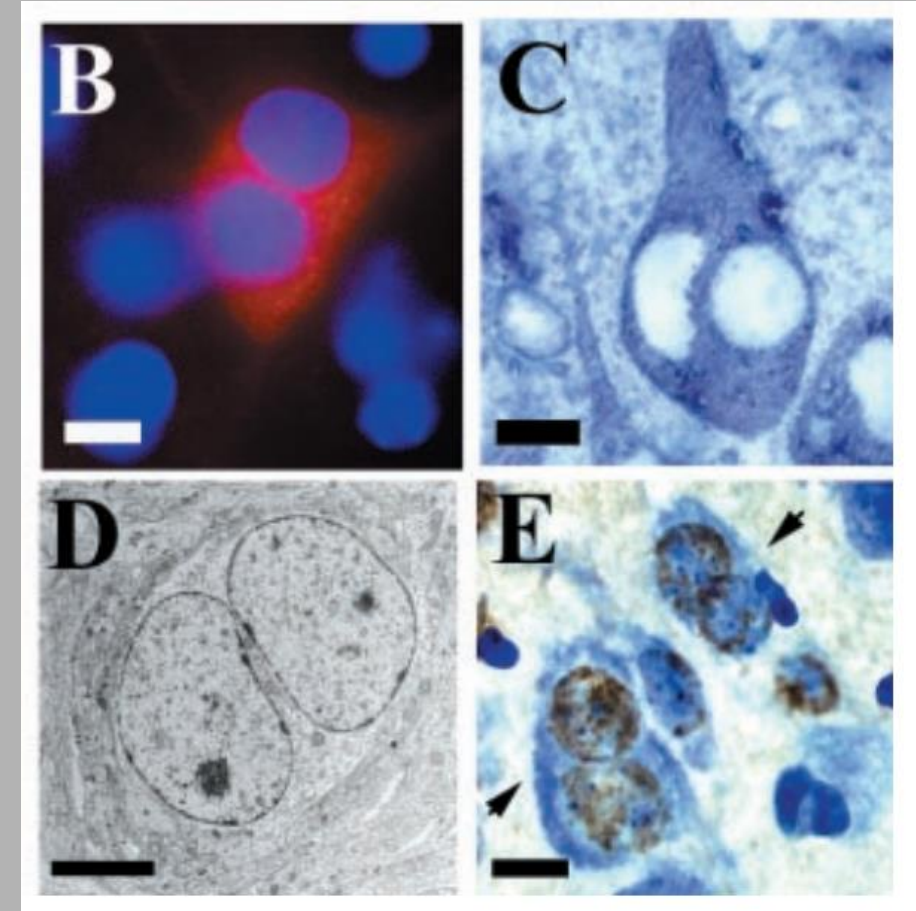
## Cytokinesis



**Citron Kinase mutations cause microcephaly by impairing cytokinesis in radial glia resulting in binucleate neurons and a reduction in the total number of neurons in brain.**



Harding BN, et al. Am J Hum Genet. 2016 Aug 4;99(2):511-20.



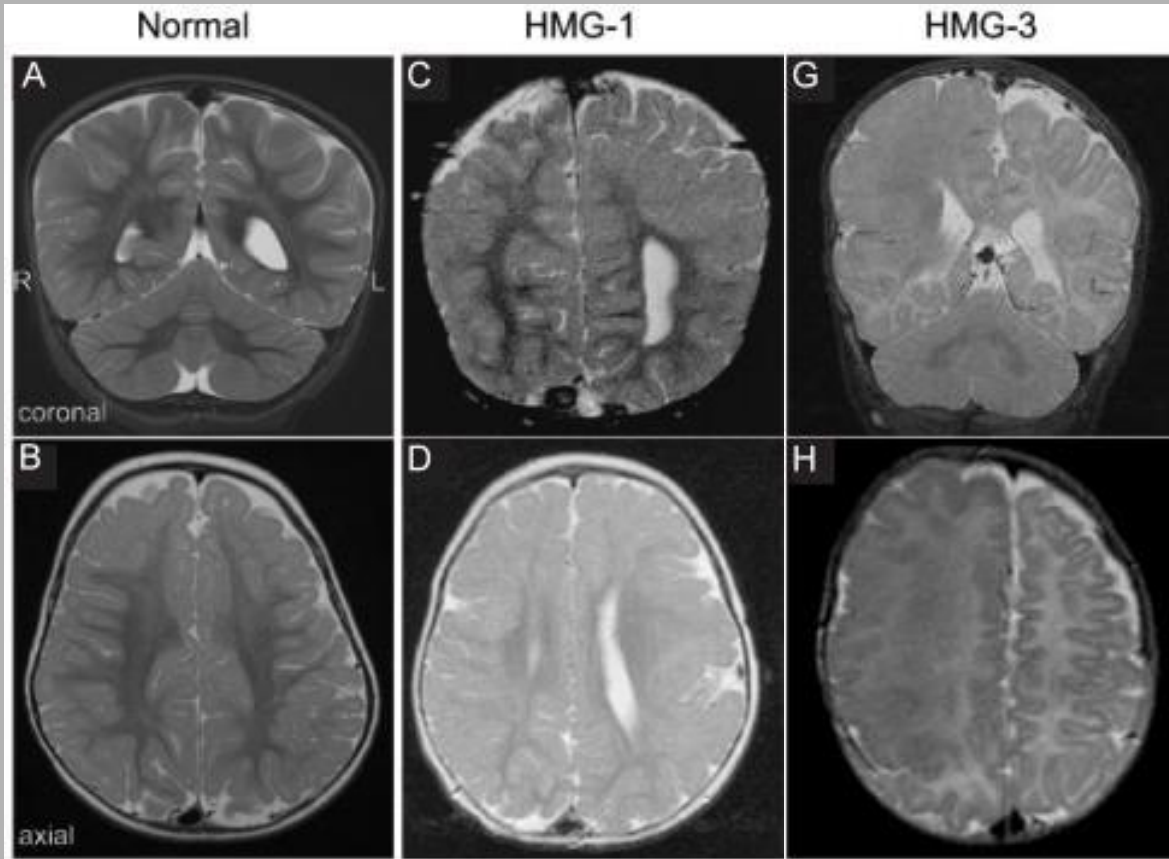
Sarkisian et al. 2002; J Neuroscience



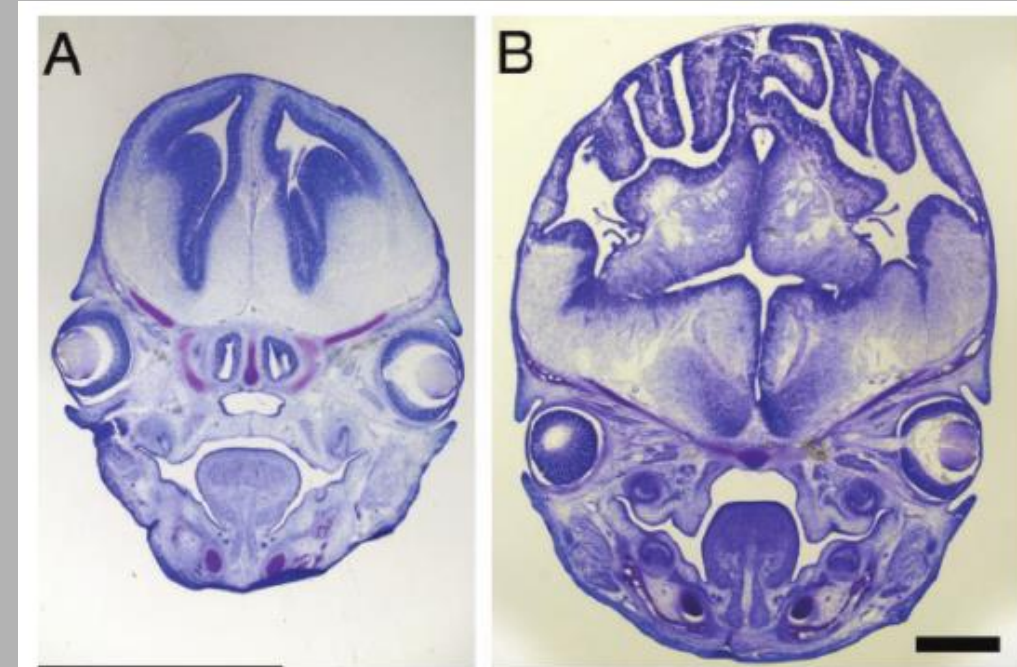
# Why is it important to understand neuronal proliferation in the brain?

## Neuronal production disorders: *Hemi-megalecephaly*

Poduri et al. Neuron. 2012 Apr 12; 74(1): 41–48.



Experimental megalecephaly by increasing proliferation in radial glia.

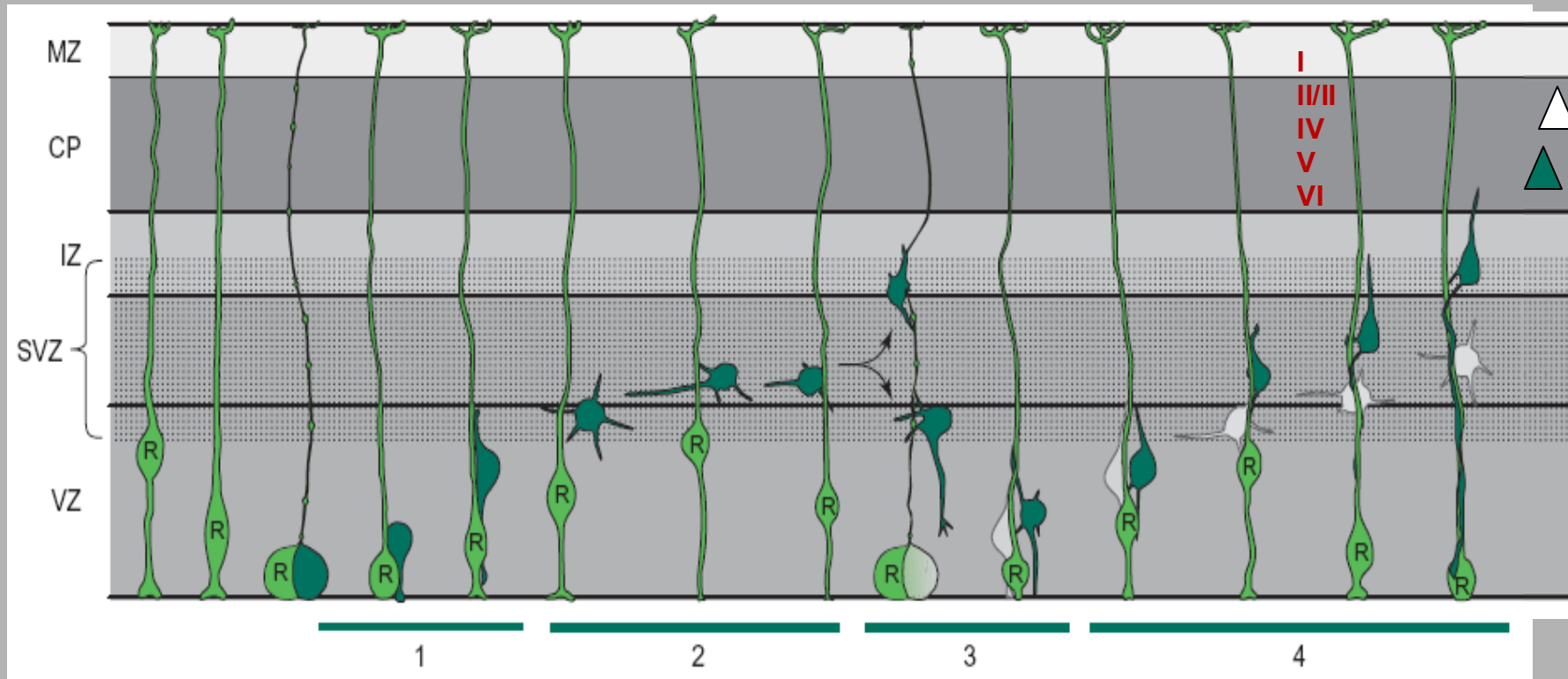


Beta catenin /nestin over exp;  
Chenn A & Walsh CA 2002

\* Enlarged and malformed hemisphere with associated epilepsy and cognitive delay/disability  
Genetic studies show - activating mutation in *AKT3* which causes prolonged proliferative capacity of radial glia. *AKT3* expression found in radial glia at VZ



# Neuronal placement includes dynamic cellular movement and dramatic changes in morphology



**SHAPE:** Bipolar  
**MVMNT:** Radial

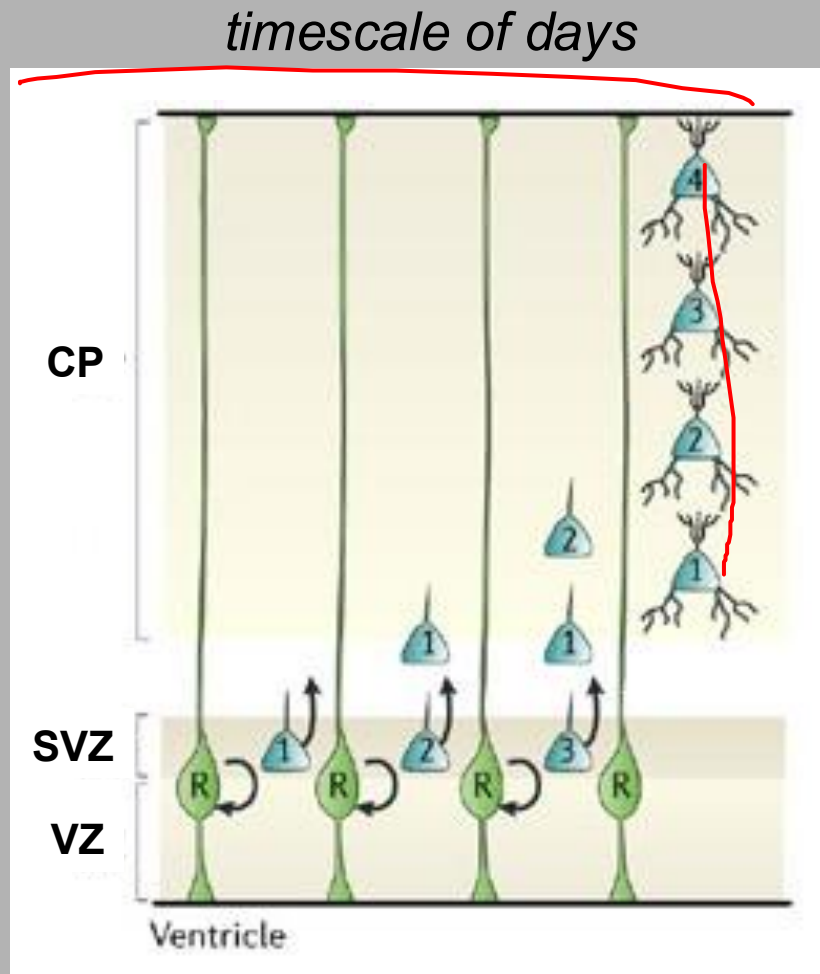
**SHAPE:** Multipolar  
**MVMNT:** Dynamic

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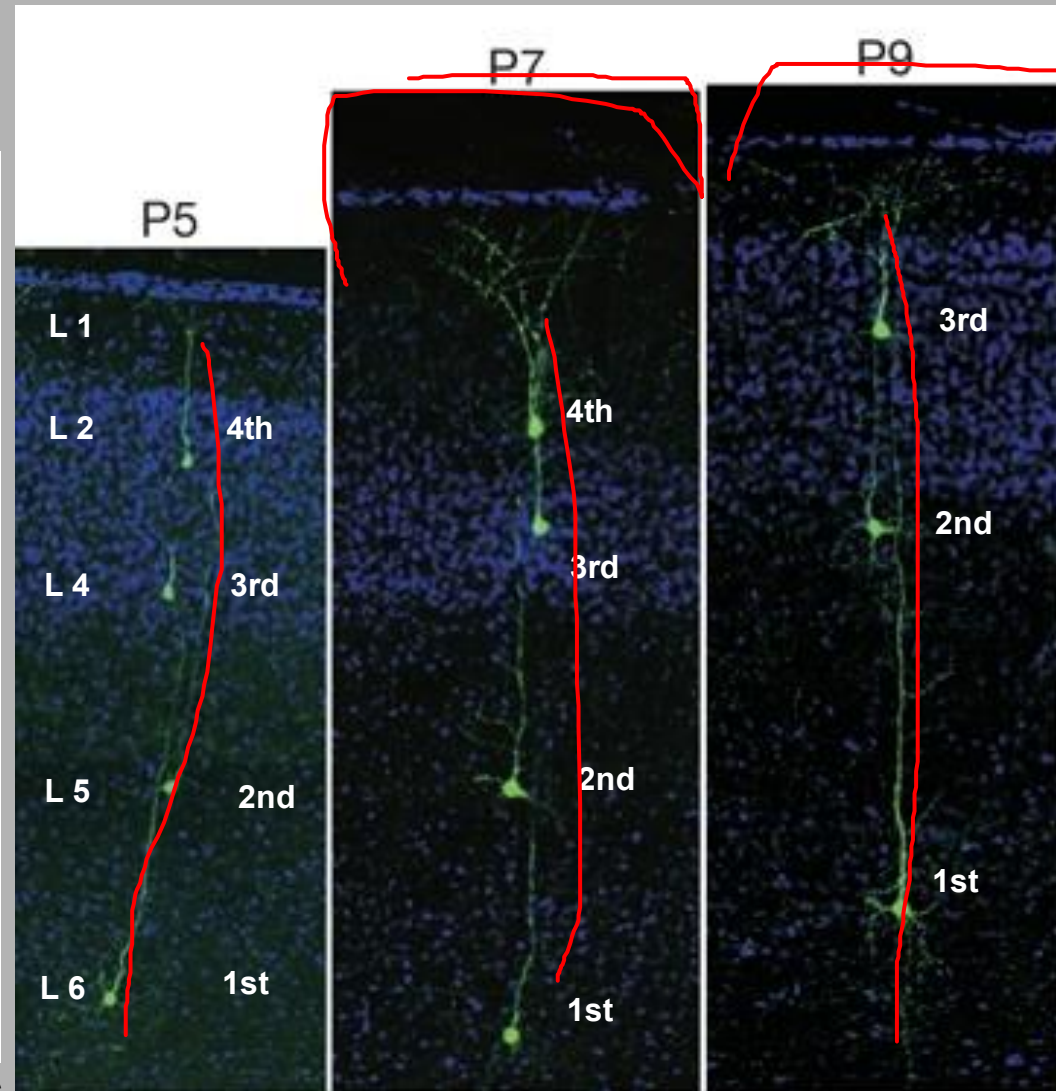
**SHAPE:** Bipolar  
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# Sequential neuronal proliferation and inside-out radial migration establishes neocortical lamina



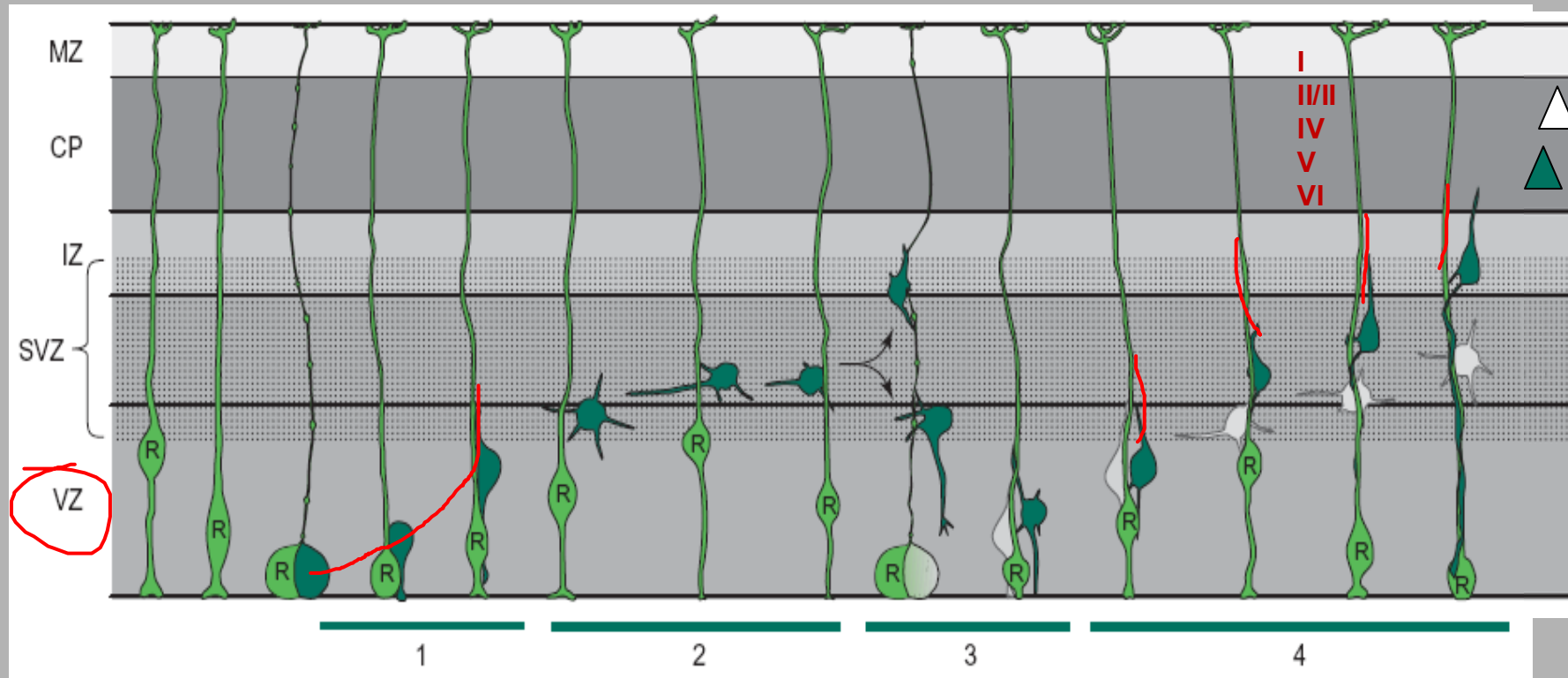
Kriegstein et al. 2006



*Nature* 458, 501-504 (2009)



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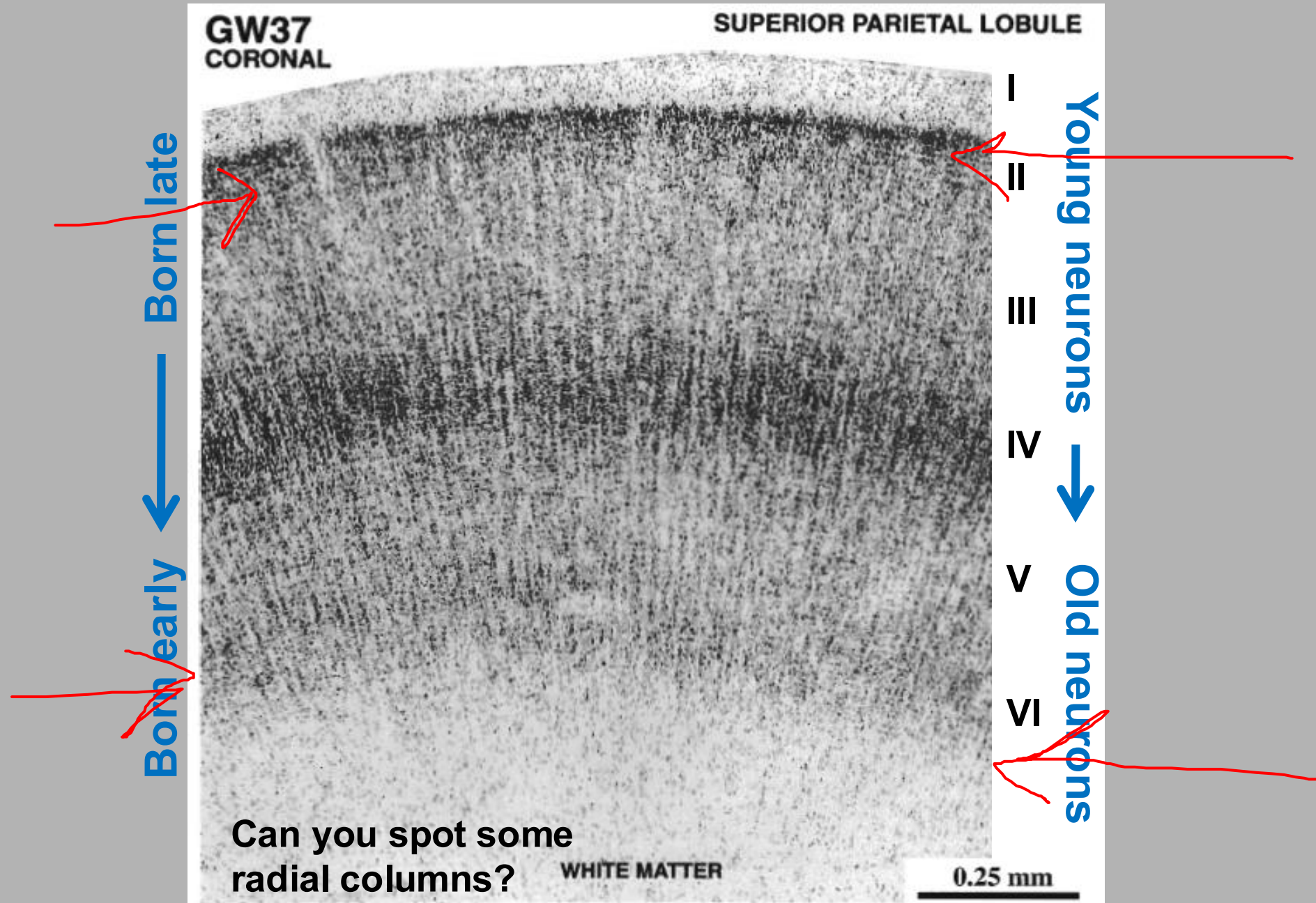
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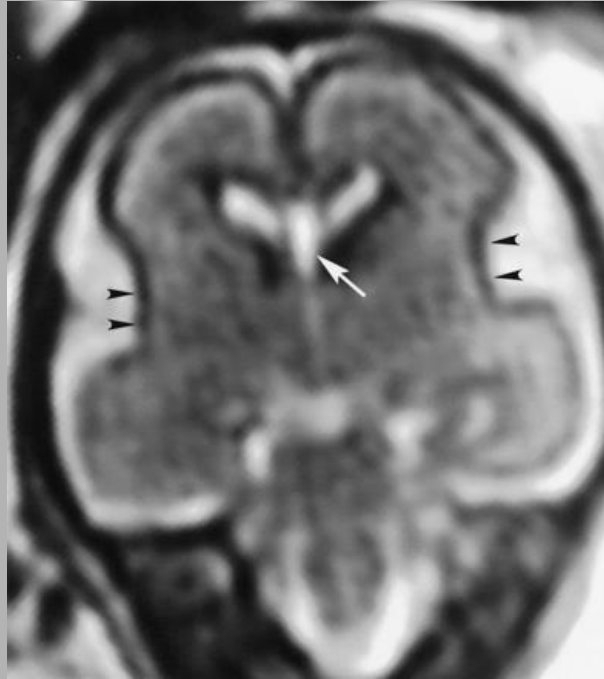


# *Inside-out radial migration establishes neocortical lamination*

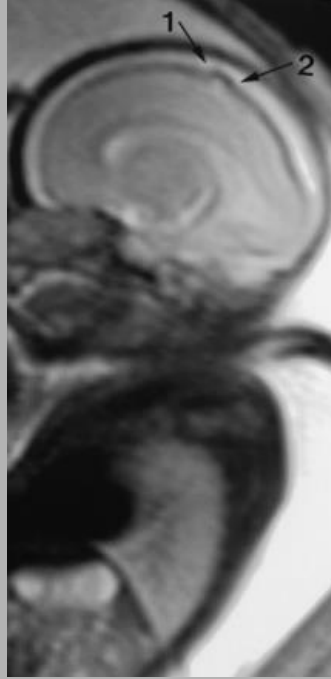


# Development of neocortical sulci & gyri

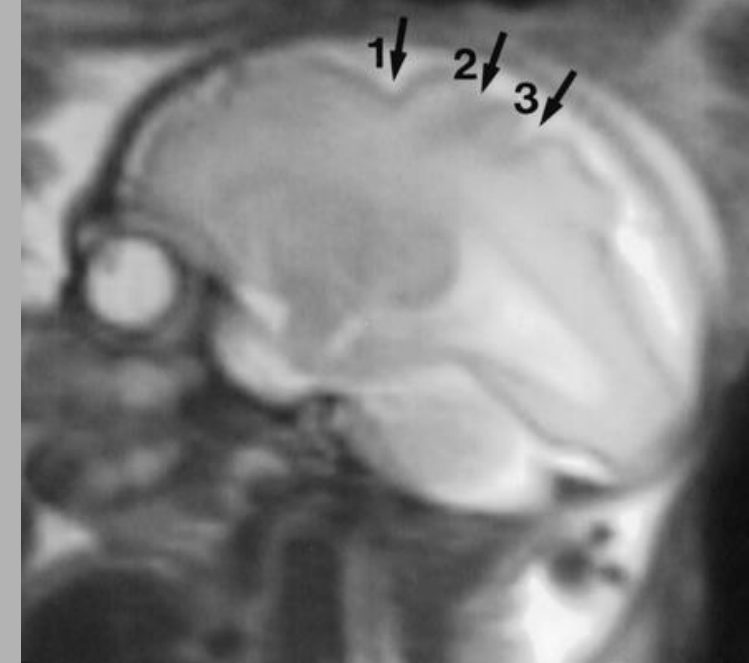
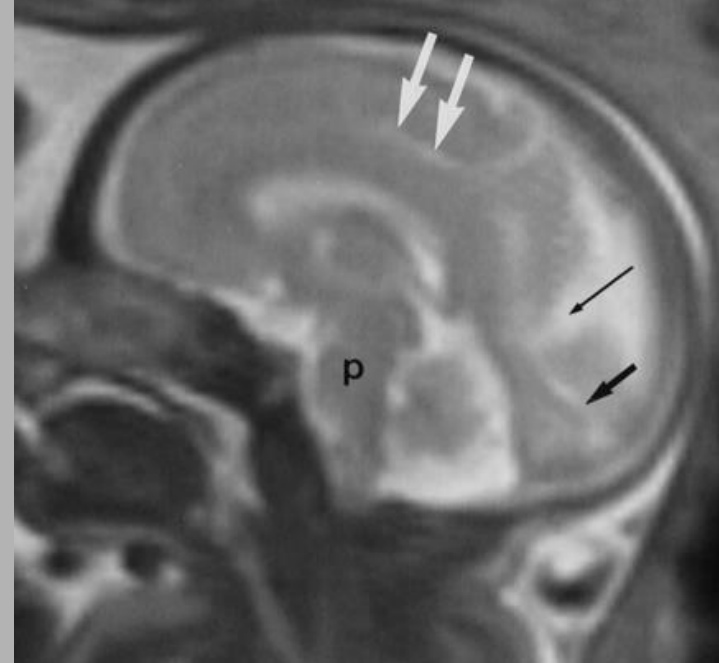
18 WEEKS



26 WEEKS



30 WEEKS



The sylvian fissure begins as a shallow depression at 14 weeks gestation and subsequently becomes grooved.

Early appearance of the precentral (arrow 1) and central (arrow 2) sulci, is depicted.

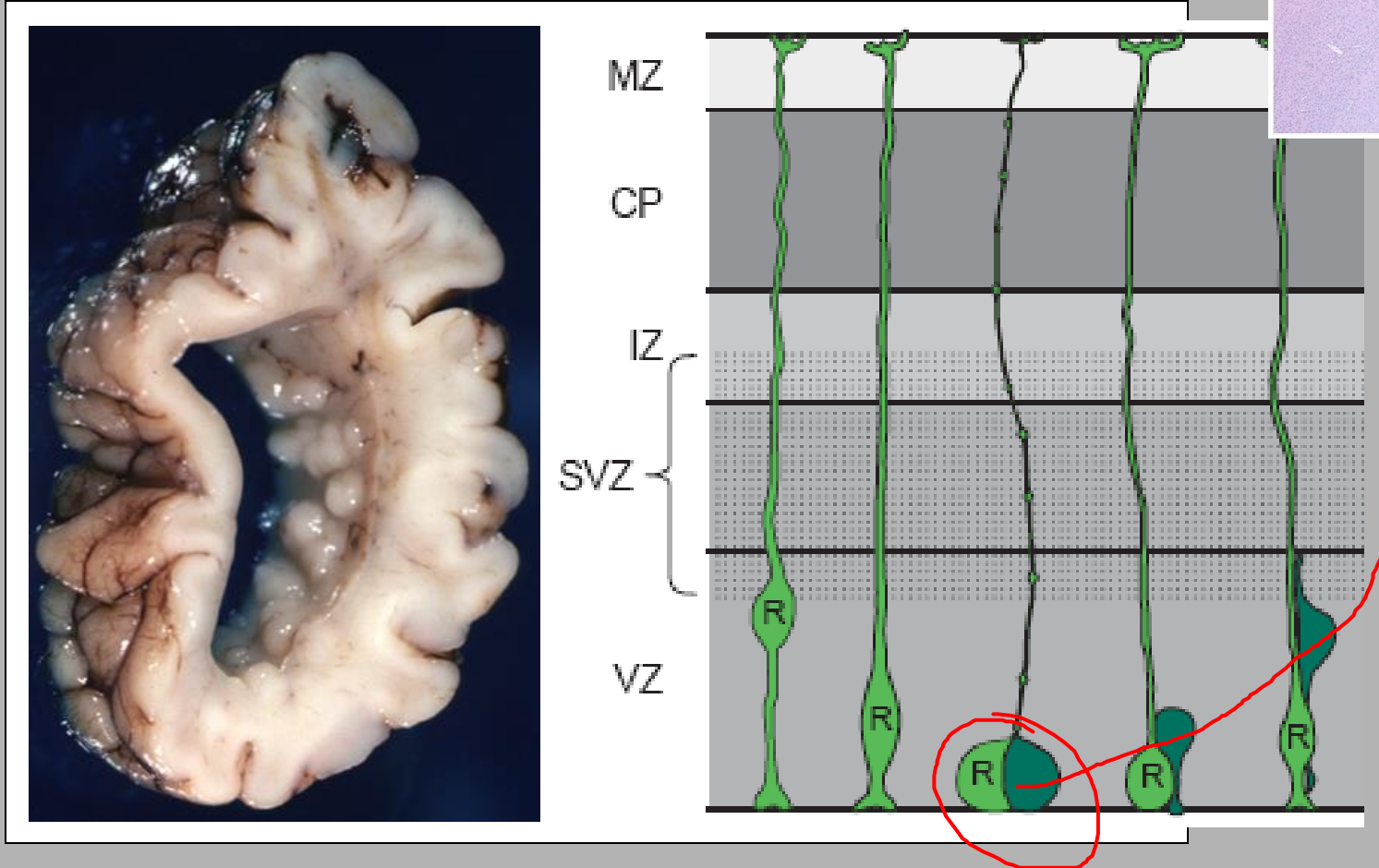
Left, The cingulate sulcus (white arrows), parieto-occipital fissure (thin black arrow), and calcarine fissure (thick black arrow) are seen. *p* = pons. Right, the precentral (arrow 1), central (arrow 2), and postcentral (arrow 3) sulci are seen.



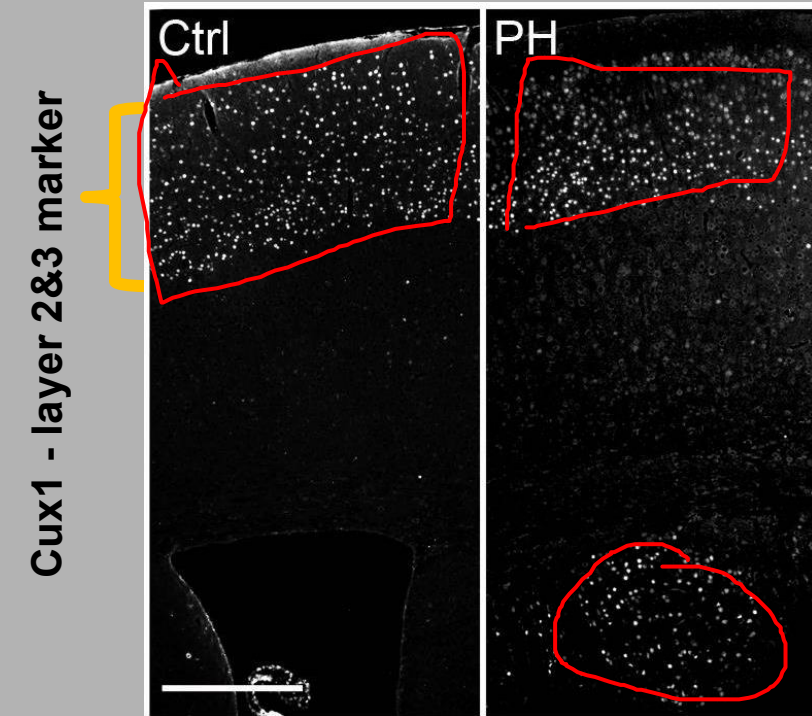
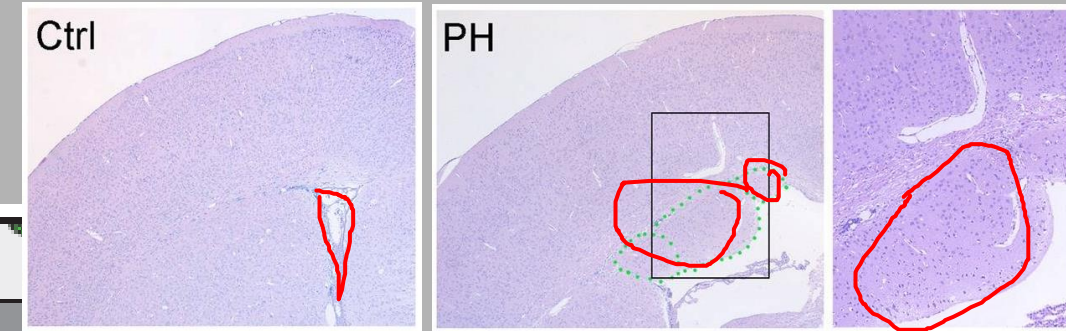
# Neuronal placement disorders:

## *Periventricular-nodular heterotopia (PVH)*

- neurons fail to leave the ventricular zone forming nodules



### *PVH in mice w/ *Flna* mutation*



- associated with mutations of Filamin A (*Flna*) gene – actin binding protein
- accompanied by cognitive impairment and epilepsy

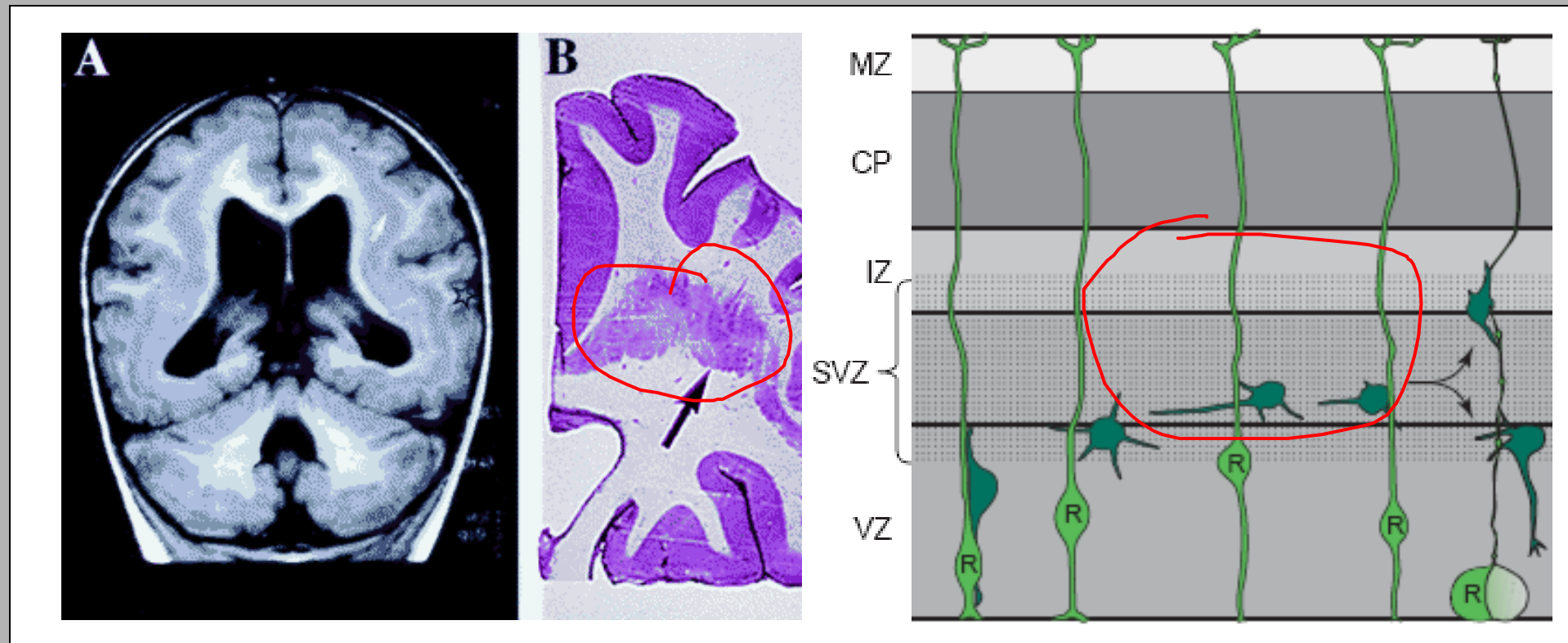
Houlihan SL et al. 2016



## Neuronal placement disorders:

### ***Subcortical Band Heterotopia/ “Double-cortex syndrome”***

- *neurons stuck in the IZ which becomes the white matter. Heterotopia are surrounded by WM*
- *associated with mutations of the EML1, DCX, LIS1 – all microtubule associated proteins*
- *accompanied by cognitive impairment and epilepsy*

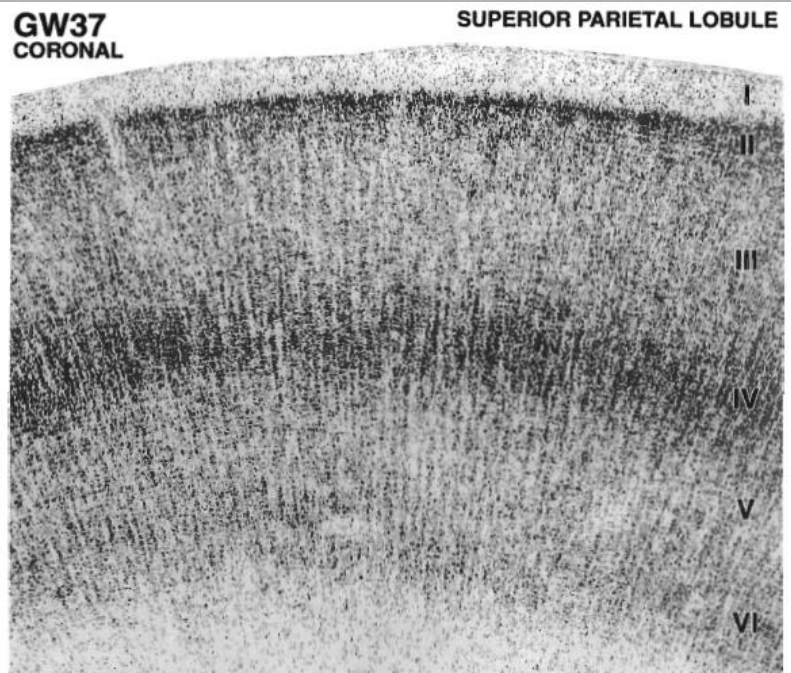


- *neurons fail to leave the intermediate zone (which becomes the WM)*



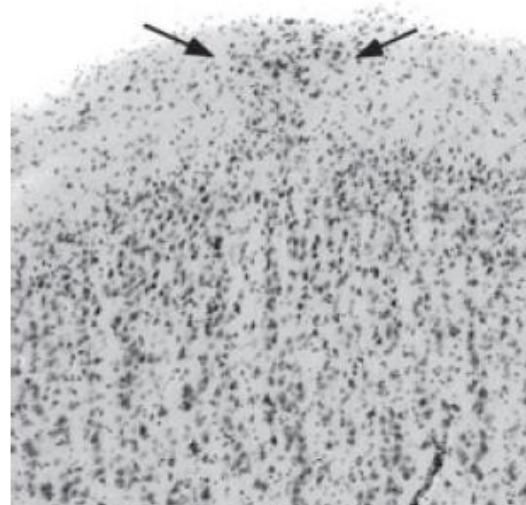
# Neuronal placement deficits can result in *Molecular-layer heterotopia*

- *neurons migrate too far into the molecular layer*

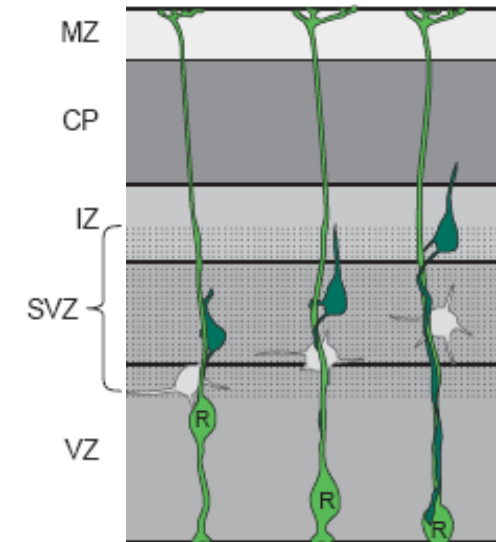


Molecular Layer  
aka Layer I

Layer 2/3

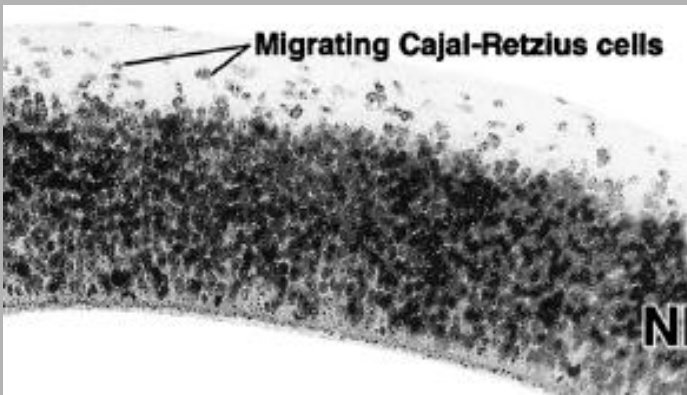


Geschwind 1990



Found in brains of epilepsy patients and dyslexic patients.





## Lissencephaly due to Reelin mutations:

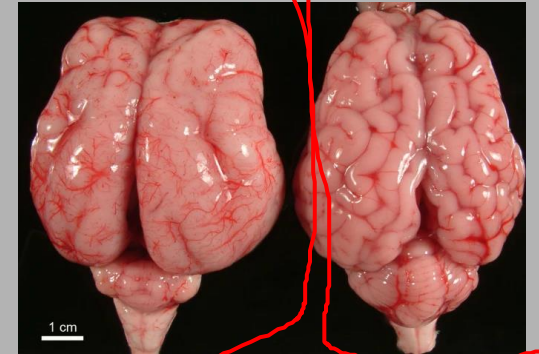
Cajal-Retzius cells are found in the marginal zone of the developing neocortex and secrete reelin. Reelin is an important molecular cue that guides neurons to their appropriate layer.

## Sheep w mutant Reelin gene

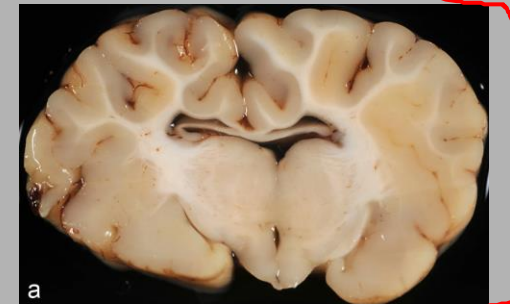


**Mutant**

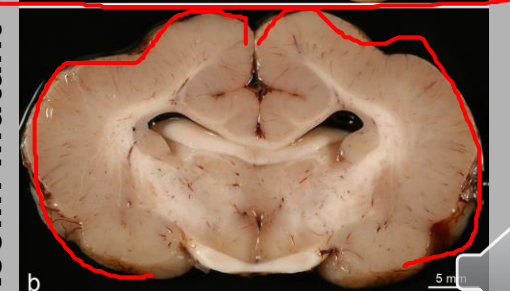
**Normal**



**Normal**



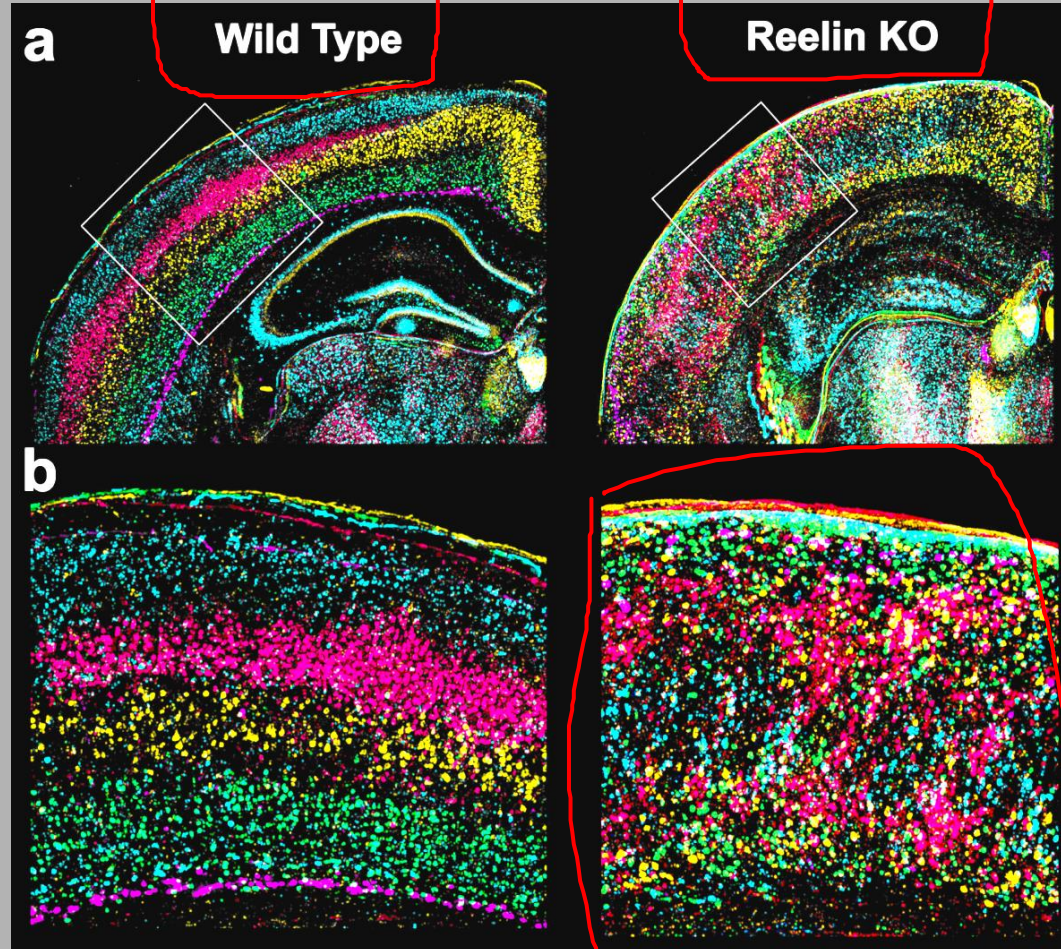
**Reelin mutant**



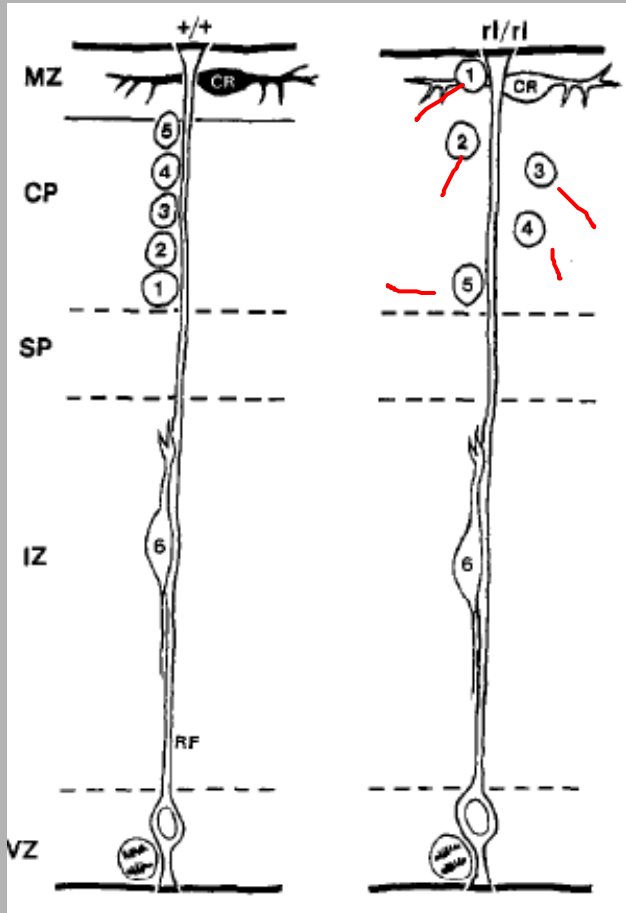
## Mouse with mutant Reelin gene

**Wild Type**

**Reelin KO**



Boyle et al. 2011



Rakic & Caviness 1995

Perez et al. 2013